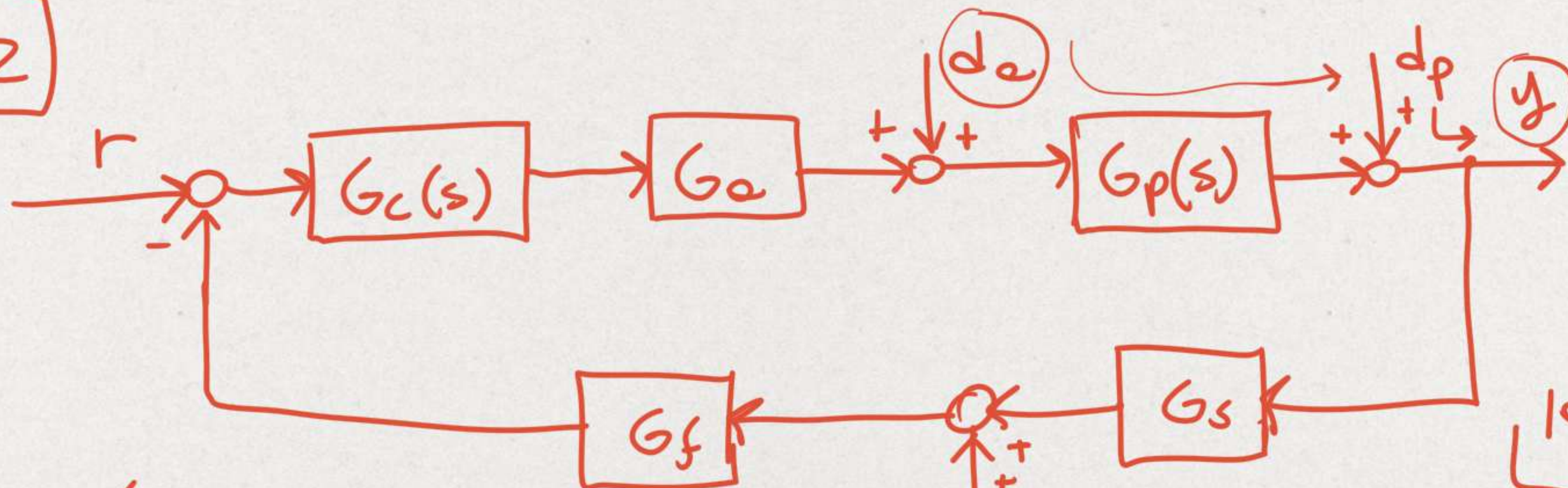


Problem 2



$$G_p(s) = \frac{40}{s^2 + 3s + 4.5}$$

$$G_s = 1; \quad G_e = -0.09$$

$$d_e(t) = D_{e\phi} \cdot \varepsilon(t), \quad |D_{e\phi}| \leq 8.5 \cdot 10^{-3}$$

$$d_p(t) = D_{p\phi} \cdot t \cdot \varepsilon(t), \quad |D_{p\phi}| \leq 3 \cdot 10^{-3}$$

$$|a_s| \leq 0.01$$

$$\omega_s \geq 50 \text{ rad/s}$$

$$d_s(t) = a_s \cdot \sin(\omega_s t)$$

• Requirements !

- ① $k_d = 1$
 - ② $|e_r^\infty| \leq 3.5 \cdot 10^{-1}$
 - ③ $|e_{da}^\infty| \leq 1.75 \cdot 10^{-2}$
 - ④ $|e_{dp}^\infty| \leq 1 \cdot 10^{-3}$
 - ⑤ $|e_{ds}^\infty| \leq 2 \cdot 10^{-4}$
- $r(t) = t \cdot \varepsilon(t)$

- ⑥ $t_r \leq 2.5s$
- ⑦ $t_{s, 5\%} \leq 10s$
- ⑧ $\hat{s}_{\%} \leq 8\%$

$$\textcircled{1} K_d = 1 \longrightarrow K_d = \frac{1}{G_f G_s} \Rightarrow \boxed{G_f = \frac{1}{K_d G_s} = 1}$$

$\uparrow$ $\underbrace{\nu + p > \phi}$ $\leftarrow$ $\uparrow$ $p = \phi \Rightarrow \nu \geq 1? \Rightarrow \boxed{\nu = 1}$

$$\textcircled{2} |e_r^\infty| \leq 0.35$$

$\hookrightarrow$ Table on slide part 7-44 $\Rightarrow \nu + p = 1 \Rightarrow \boxed{\nu = 1 - p = 1}$

$$\Rightarrow |e_r^\infty| = \left| \frac{K_d^2 \cdot R_o}{K_p \cdot K_c \cdot G_o} \right| \Rightarrow \left| \frac{1 \cdot 1}{8.89 \cdot K_c \cdot (-0.09)} \right| \leq 0.35 \Rightarrow \boxed{|K_c| \geq 3.57}$$

$\uparrow$ $K_p = \lim_{s \rightarrow 0} s^p \cdot G_p(s) = \frac{40}{4.5} \approx 8.89$

$$\textcircled{3} |e_{de}^{\infty}| \leq 1.75 \cdot 10^{-2}$$

$$|e_{de}^{\infty}| \stackrel{\Delta}{=} |y_{de}^{\infty}| \stackrel{\Delta}{=} \left| \lim_{t \rightarrow \infty} y_{de}(t) \right| = \left| \lim_{s \rightarrow 0} s \cdot y_{de}(s) \right| = \left| \lim_{s \rightarrow 0} s \cdot G_{de}(s) \cdot d_e(s) \right| =$$

$$= \left| \lim_{s \rightarrow 0} s \cdot \frac{G_p}{1+L} \cdot \frac{D_{e\phi}}{s} \right| = \left| \lim_{s \rightarrow 0} \frac{8.89 \cdot 8.5 \cdot 10^{-3}}{1 + \frac{K_L}{s^v} \cdot 8.89 \cdot (-0.09) \cdot 1 \cdot 1} \right| =$$

$$s \rightarrow 0 \Rightarrow \begin{cases} G_p \rightarrow \frac{K_p}{s^p} = K_p = 8.89 \\ G_L \rightarrow \frac{K_L}{s^v} \end{cases}$$

$$= \left| \lim_{s \rightarrow 0} \frac{s^v \cdot 8.89 \cdot 8.5 \cdot 10^{-3}}{s^v + K_L \cdot 8.89 \cdot (-0.09)} \right| \leq 1.75 \cdot 10^{-2} \Rightarrow \boxed{v = \phi} \Rightarrow |K_L| > \sim \dots$$

However we know that from (2) $\Rightarrow \boxed{v = 1} \Rightarrow$
 $\Rightarrow$ we must $\boxed{v = 1} \rightarrow \forall |K_L|$

$$\textcircled{4} \quad |e_{dp}^\infty| \leq 1 \cdot 10^{-3}$$

$$|e_{dp}^\infty| = |y_{dp}^\infty| = \left| \lim_{t \rightarrow \infty} y_{dp}(t) \right| = \left| \lim_{s \rightarrow 0} s \cdot y_{dp}(s) \right| = \left| \lim_{s \rightarrow 0} s \cdot G_{dpy}(s) \cdot \frac{D_{p\phi}}{s^2} \right| =$$

$$= \left| \lim_{s \rightarrow 0} s \cdot \frac{1}{1+L} \cdot \frac{D_{p0}}{s^2} \right| = \left| \lim_{s \rightarrow 0} \frac{D_{p0}}{1 + \frac{K_C}{s^V} \cdot 8.89 \cdot (-0.09)} \cdot \frac{1}{s} \right| =$$

$$s \rightarrow 0 \begin{cases} G_P \rightarrow \frac{K_P}{s^P} = K_P = 8.89 \\ G_C \rightarrow \frac{K_C}{s^V} \end{cases}$$

$$\boxed{|K_C| \geq 3.75}$$

$\Rightarrow$

$$= \left| \lim_{s \rightarrow 0} \frac{\textcircled{s^V} \cdot D_{p\phi}}{s^V + K_C \cdot 8.89 \cdot (-0.09)} \cdot \frac{1}{\textcircled{s}} \right| \leq 1 \cdot 10^{-3} \Rightarrow \boxed{V=1} \Rightarrow \left| \frac{3 \cdot 10^{-3}}{K_C \cdot 8.89 \cdot (-0.09)} \right| \leq 10^{-3}$$

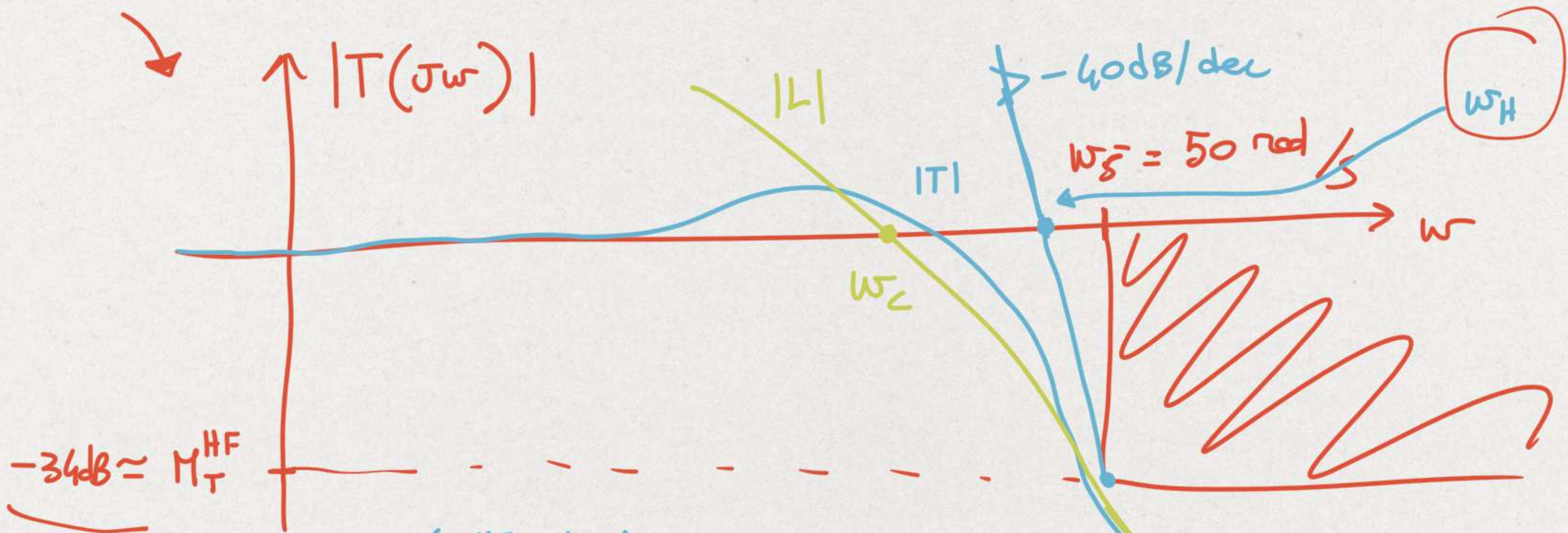
$$\textcircled{5} \quad |e_{ds}^\infty| \leq 2 \cdot 10^{-4}$$

$$|e_{ds}^\infty| \triangleq |y_{ds}^\infty| = |a_s| \cdot |G_{dsy}(j\omega_s)| \cdot \overbrace{\sin(\omega_s \cdot t + \varphi)}^{\in [-1, 1]} \leq$$

$$\leq |a_s| \cdot |G_{dsy}(j\omega_s)| \leq |a_s| \cdot |T(j\omega_s)| \cdot \frac{1}{G_s} \leq 2 \cdot 10^{-4} \Rightarrow$$

$$G_{dsy}(s) = T(s) \cdot \frac{1}{G_s}$$

$$\Rightarrow |T(j\omega_s)| \leq \frac{2 \cdot 10^{-4} \cdot 1}{10^{-2}} = M_T^{HF} \rightarrow \simeq -34 \text{ dB} \quad \forall \omega_s \geq 50 \text{ rad/s}$$



$$\omega_H = \omega_s \cdot 10^{(M_{\text{TdB}}^{\text{HF}} / 40)}$$

$$\omega_c \leq \frac{\omega_H}{2} \approx 3.5 \text{ rad/s}$$

- ⑥ $t_r \leq 2.5s$
- ⑦ $t_{s,5\%} \leq 10s$
- ⑧ $\hat{s}\% \leq 8\%$

$\Downarrow$
 $\hat{s}\% \leq 8\% \Rightarrow \hat{s} = e^{\frac{-\pi \xi}{\sqrt{1-\xi^2}}} \Rightarrow \xi = \frac{|\ln(\hat{s})|}{\sqrt{\pi^2 + \ln^2(\hat{s})}} \approx 0.63$

(slide Port 7-116)

$\hat{s} = 0.08$

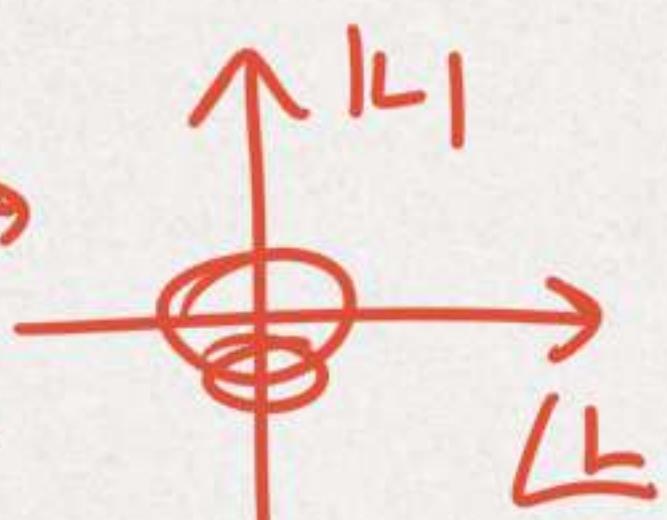
$\hat{s} \leq 0.08 \Rightarrow \xi \geq 0.63 \Rightarrow$

Slide Port 7-118

$T_{p_{\max}} = \frac{1}{2\xi \cdot \sqrt{1-\xi^2}} \approx 1.024$

$S_{p_{\max}} = \frac{2\xi \sqrt{2+4\xi^2} + 2 \cdot \sqrt{1+8\xi^2}}{\sqrt{1+8\xi^2} + 4\xi^2 - 1} \approx 1.33$

⑧ $\Rightarrow \zeta \geq 0.63 \rightarrow \left. \begin{array}{l} T_p \leq 1.024 \\ S_p \leq 1.33 \end{array} \right\} \rightarrow \text{myrqidst}(T_{pmax}, S_{pmax}) \rightarrow$



⑥ $t_r \leq 2.5 \text{ s}$

(slide part 7-121) $\Rightarrow t_r \cdot \omega_c = \frac{1}{\sqrt{1-\zeta^2}} \cdot (\pi - \arccos(\zeta)) \cdot$

$\cdot \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2} \Rightarrow \boxed{\omega_c \geq 0.81 \text{ rad/s}}$

$\uparrow$
 $t_r \leq 2.5 \text{ s}$

$$\textcircled{7} \quad t_{s, 5\%} \leq 10s$$

(slide part 7-123)

$$t_{s, \alpha\%} \cdot \omega_c = \frac{-\ln(\alpha)}{\zeta} \cdot \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

$$\zeta = 0.63$$

$$\omega_c \geq 0.33 \text{ rad/s}$$

$$t_{s, 5\%} \leq 10s$$



Summary of obtained design constraints :

① $G_f = 1$

② $V = 1$, $|K_c| \geq 3.57$

③ $V = 1$, $\forall |K_c|$

④ $V = 1$, $|K_c| \geq 3.75$

⑤ $\omega_c \leq 3.5 \text{ rad/s}$

⑥ $\omega_c \geq 0.81 \text{ rad/s}$

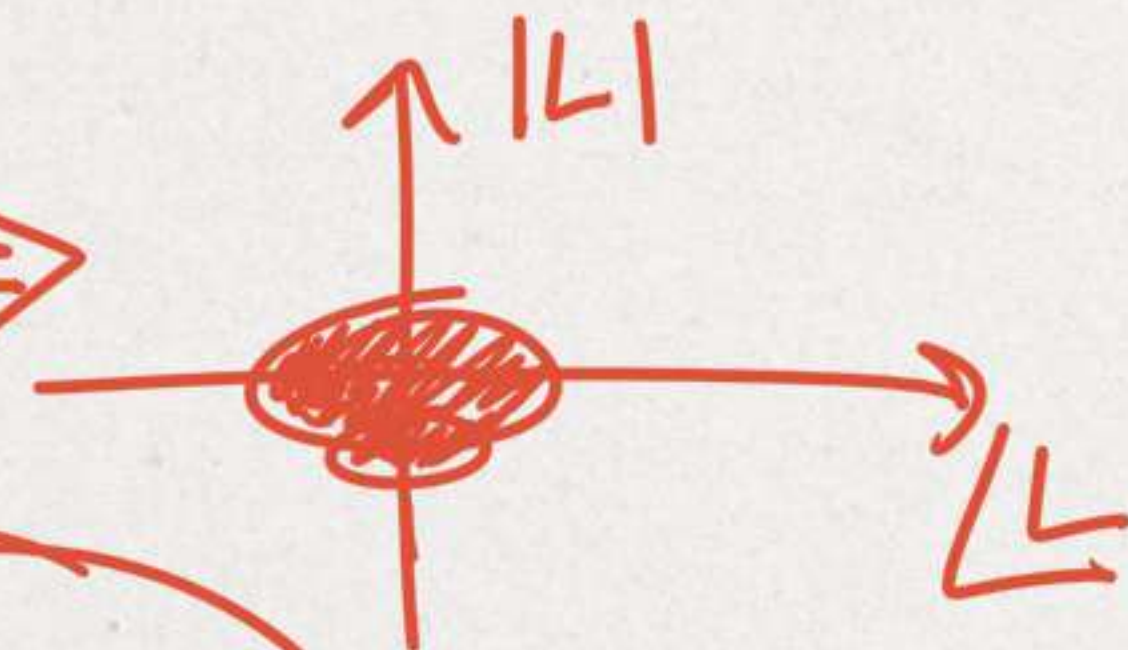
⑦ $\omega_c \geq 0.33 \text{ rad/s}$

$V = 1$

$|K_c| \geq 3.75$

⑧ $\Rightarrow$

$|K_c| = 3.8$



$\Rightarrow \omega_c \in [0.81, 3.5] \text{ rad/s}$

Supponiamo che in un dato generico problema si
trovi!

I

$$v = 1, \quad |k_c| > 7$$

II

$$v = 1, \quad \forall k_c$$

II

$$v = 2, \quad |k_c| > 3$$

IV

$$v = 2, \quad |k_c| > 0.1$$

$$\boxed{v = 2}, \quad |k_c| > 3$$

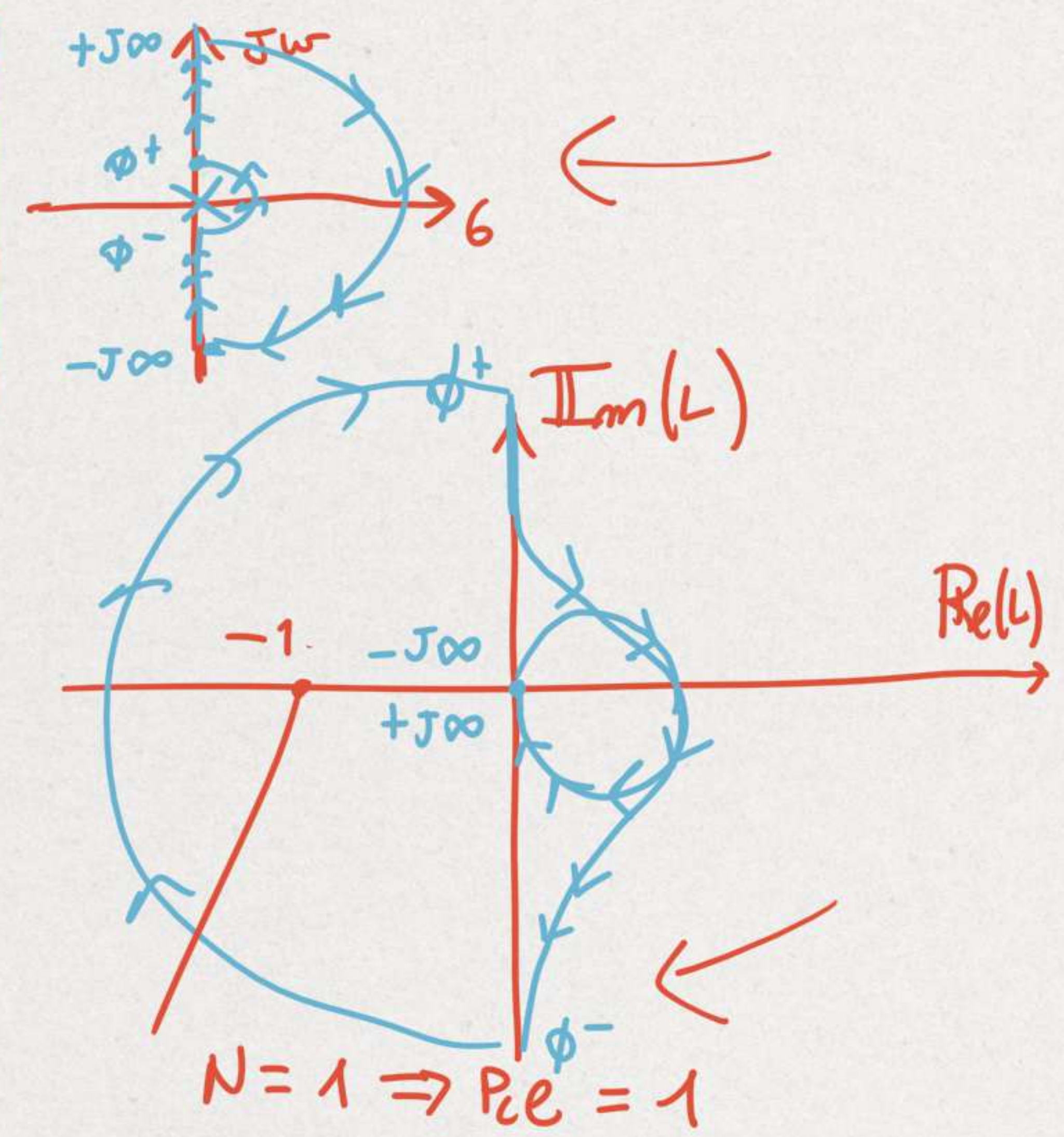
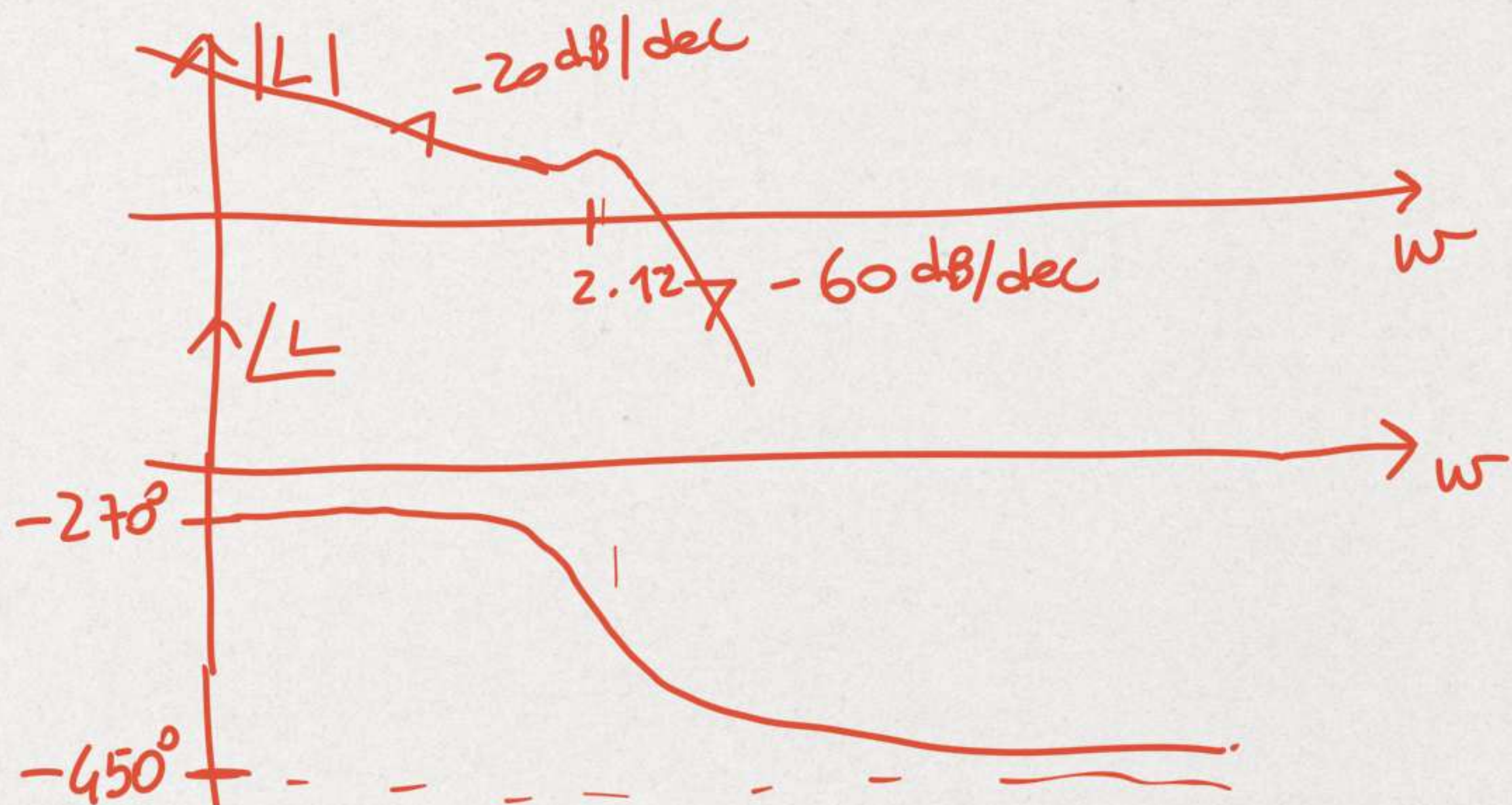
Design of the controller $G_C(s)$

First step! selection of the sign of K_C

Assume that $K_C > 0 \Rightarrow G_C(s) = \frac{3.8}{s} \Rightarrow$

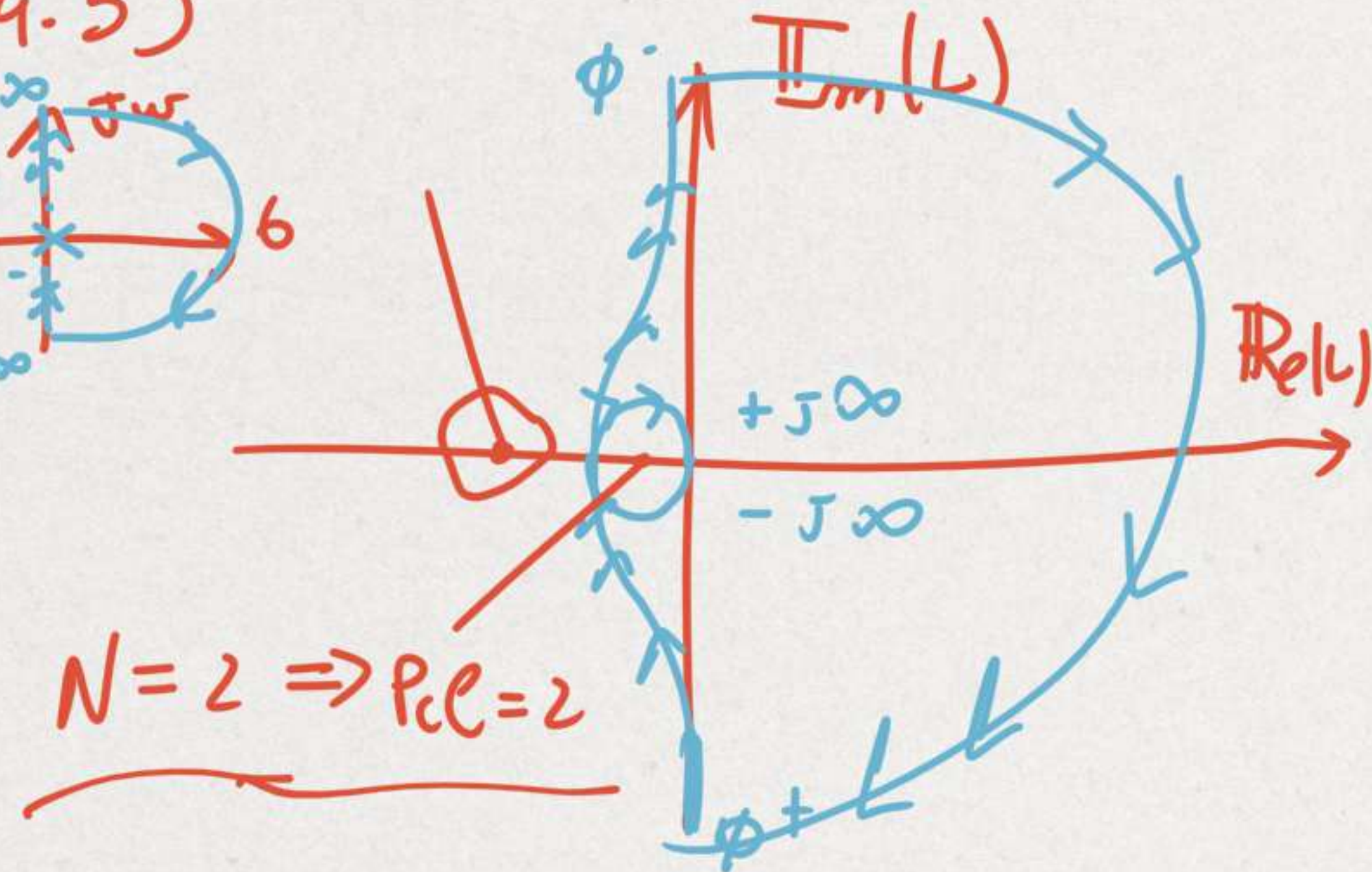
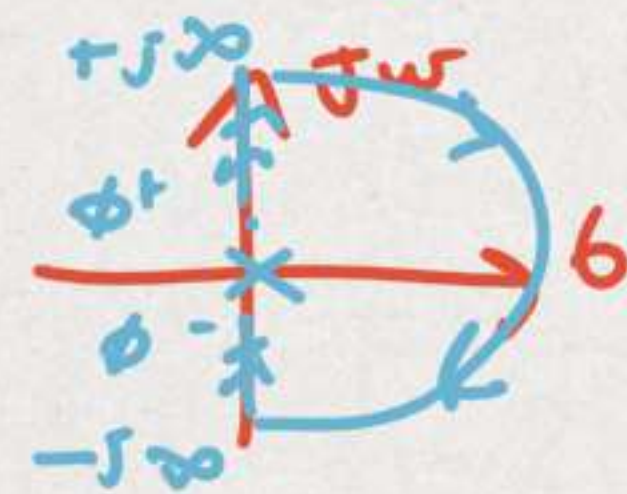
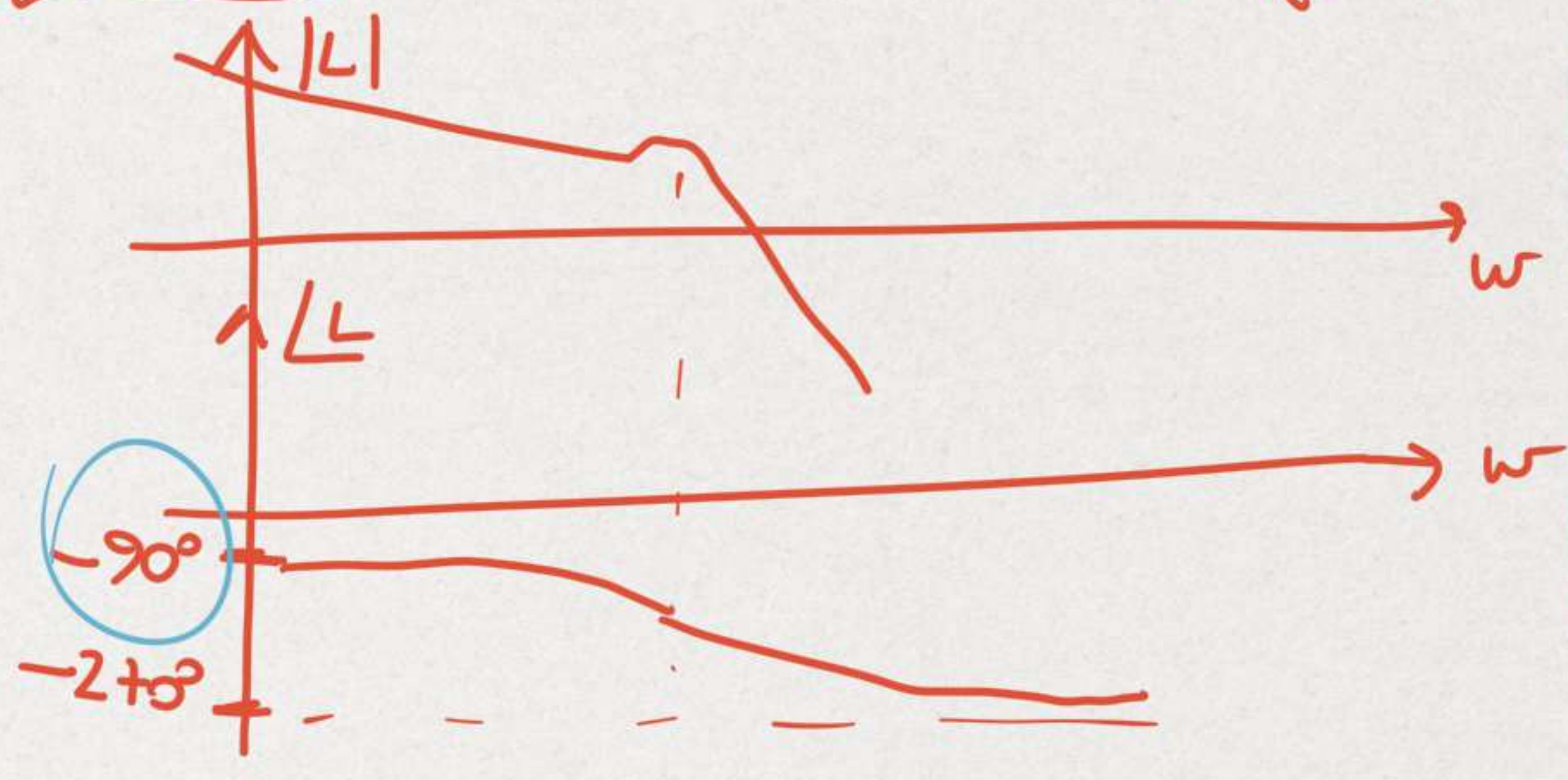
$$\Rightarrow L(s) = \frac{3.8}{s} \cdot G_P(s) \cdot G_a \cdot G_s \cdot G_f = \frac{-136.8}{s(s^2 + 3s + 4.5)}$$



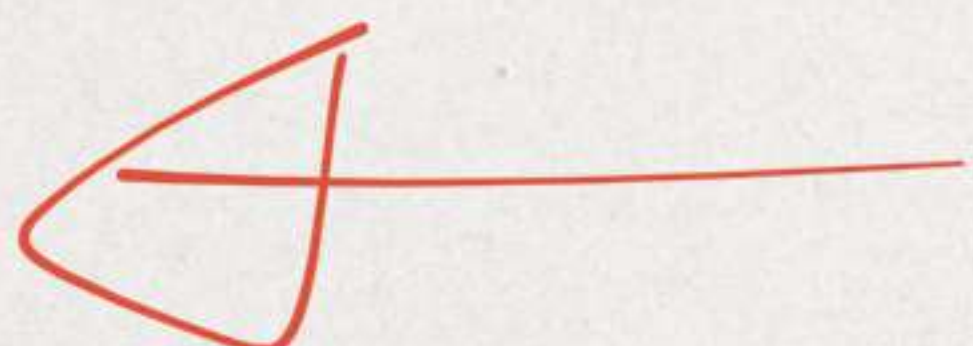


Since with $K_c > 0 \Rightarrow P_c L = 1 \Rightarrow$ we cannot stabilize the feedback system!

$\Downarrow$
 $K_c < 0 \Rightarrow L_m(s) = \frac{136.8}{s \cdot (s^2 + 3s + 4.5)}$



Since with $K_c < 0 \Rightarrow P_{cl}$ is even $\Rightarrow$ we can stabilize the feedback control system!

$\Downarrow$
So we select $K_c < 0 \Rightarrow \boxed{K_c = -3.8}$ 

Second step : selection of $\boxed{\omega_{c,des}}$

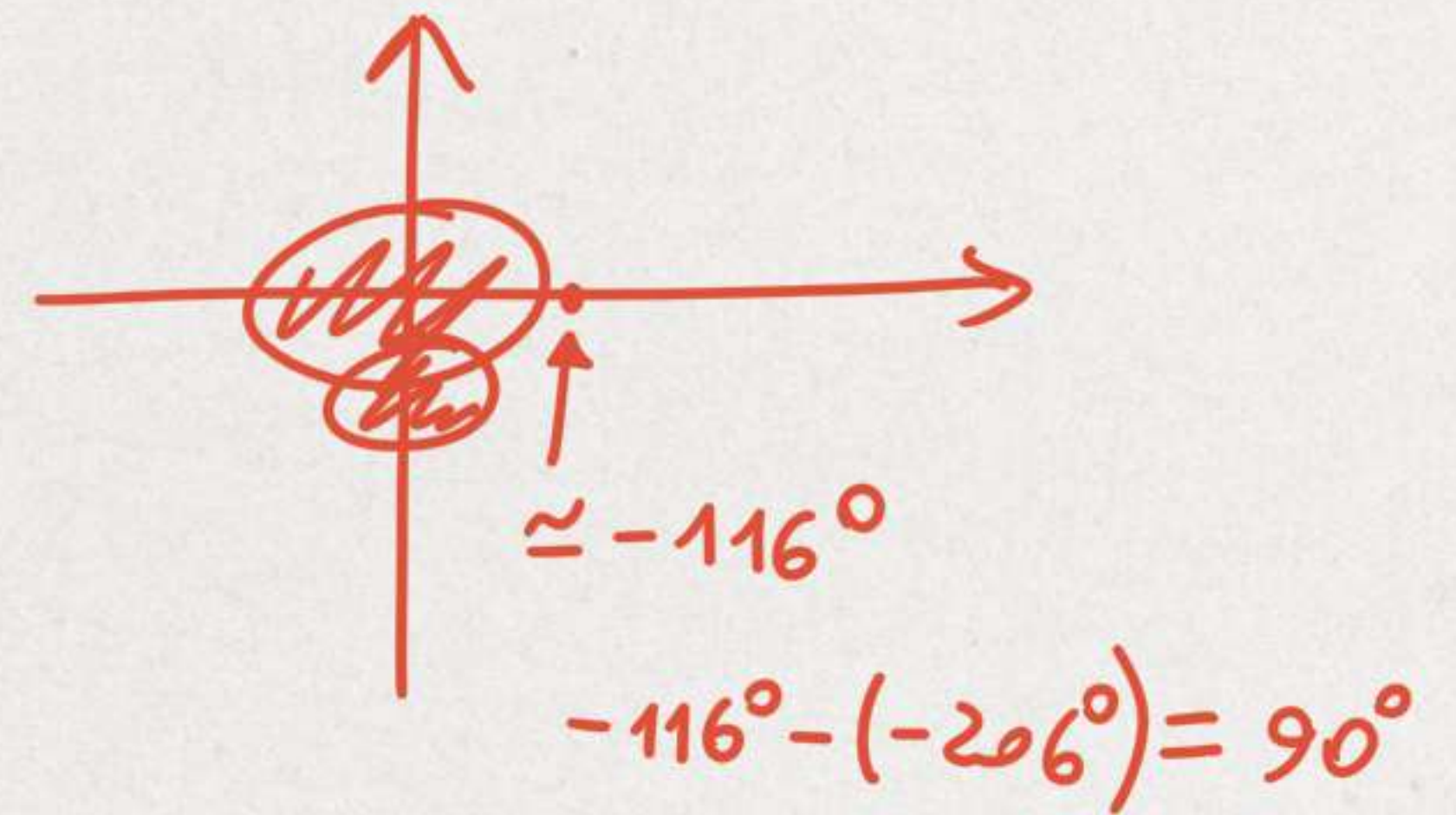
$$\omega_{c,des} \in [0.81, 3.5] \text{ rad/s}$$

By looking at the Nichol plot of L_{in} we select:

$$\omega_{c,des} = 3 \text{ rad/s}$$



$$\begin{aligned} &\Downarrow \\ |L| &\approx -6.87 \text{ dB} \\ \angle L &\approx -206^\circ \end{aligned}$$



We have to increase $|L|$ of about 7 dB and $\angle L$ of at least 90° $\nearrow$
 $\Downarrow$

- In order to increase the phase we have to insert into the controller either zero networks R_z or lead networks R_d or both
- In principle (if possible) we prefer to insert R_z because this kind of networks can provide a larger phase increase over the entire frequency range

$$R_z = \left(1 + \frac{s}{z}\right)$$



$$R_d = \frac{\left(1 + \frac{s}{z_d}\right)}{\left(1 + \frac{s}{m_d \cdot z_d}\right)}$$



- So, in principle R_z are better than R_d !
- However, we have to keep in mind that at the end of our design the obtained controller G_c must be physically realizable and, therefore, the degree of the numerator of $G_c(s)$ must always be less or equal than the degree of the denominator of $G_c(s)$!



For this reason we can add to our controller at most ν zero-poles R_z !





In particular in this problem we can insert a single R_z network since $\nu = 1!$

• Additional consideration about networks design

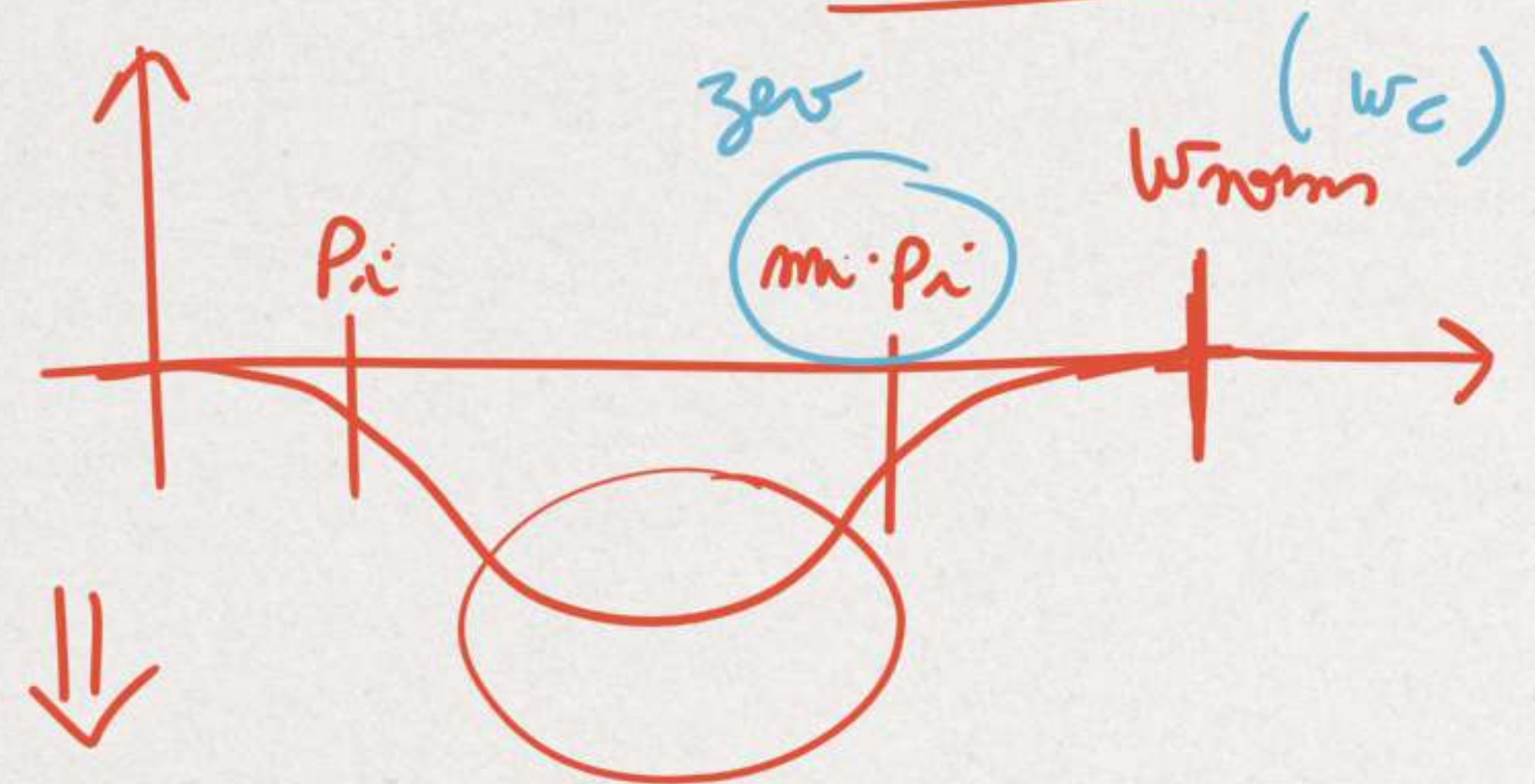
• Log network :
$$R_i = \frac{(1 + \frac{s}{\omega_i p_i})}{(1 + \frac{s}{p_i})}$$



① in the log network we have always a pole at frequency lower than the frequency of the zero

② Since, typically we select $\omega_{\text{nom}} = 100$ (or greater) for R_i (in order to move the phase lag effect far from ω_c , des)

$\Downarrow$
the pole and zero of R_i are placed typically at low frequency ($\ll \omega_c$)



③ Note that:

$$T = \frac{L}{1+L}$$

⇓

Therefore we have that
True for the poles)

and $L = \underbrace{G_c \cdot G_p \cdot G_o \cdot G_s \cdot G_f}$

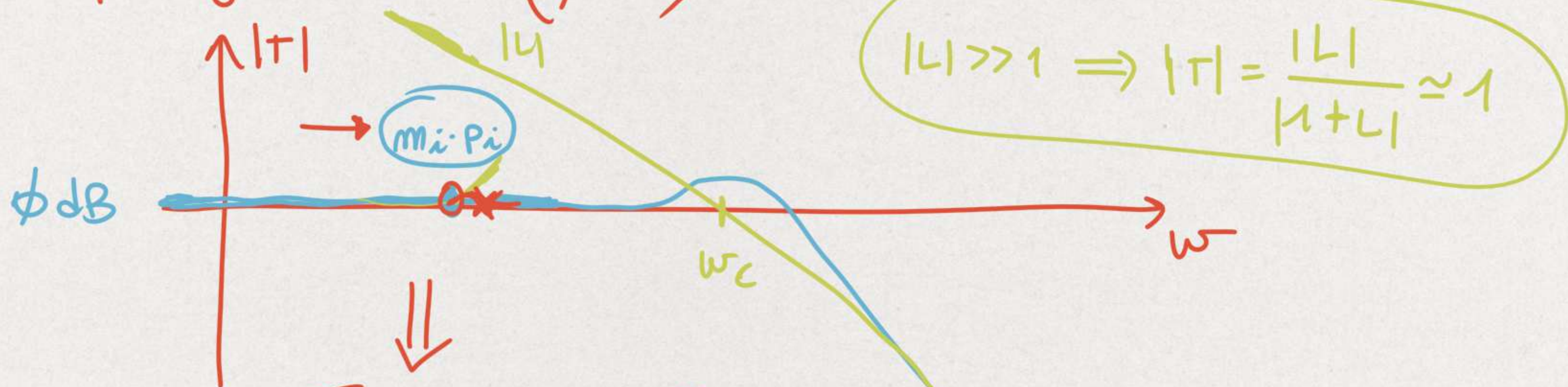
↓

$$\underbrace{\text{zeros}(L)} = \text{zeros}(T) \quad \left(\text{this is not} \right)$$

⇓

$$\text{zeros}(G_c) \text{ are also zeros}(T)$$

④ We know that at low frequency ($\ll \omega_c$) we have typically $|T|=1$ ($\phi \approx 0$)



Im order to hold $|T|=1$ ($\phi \approx 0$) in presence of the low-frequency zero of the G_f -network $\Rightarrow$ a pole at almost the same frequency is appearing in T in order to cancel $\downarrow$

→ the effect of such zero on $|T|$!



It implies that the bg -network is implicitly bringing a low-frequency pole in T

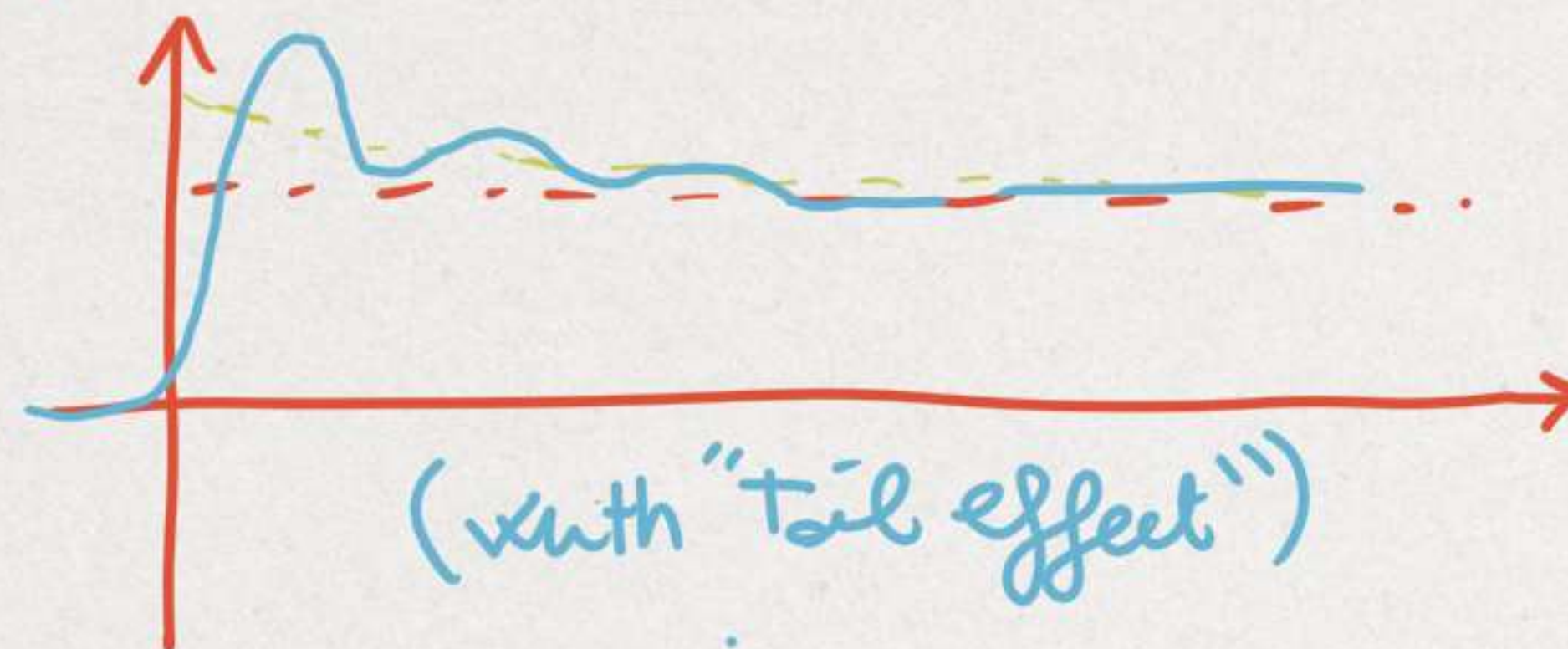
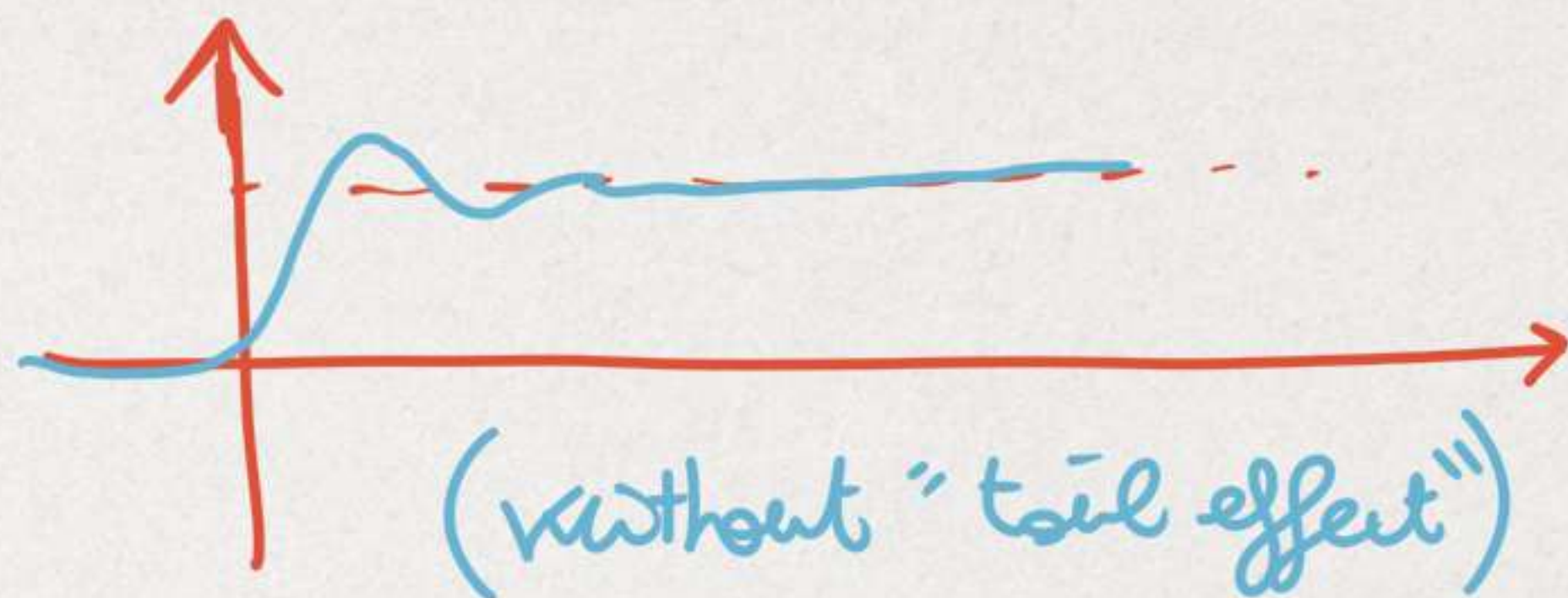
⇓
• Let's call this pole $p^* \Rightarrow y(t) = \mathcal{L}^{-1} \left\{ \overbrace{T(s) \cdot r(s)}^{y(s)} \right\} =$

$$= \mathcal{L}^{-1} \left\{ \frac{R_1}{s - p^*} + \frac{R_2}{s - p_2} + \dots \right\} =$$

→ $= \mathcal{L}^{-1} \left\{ \frac{R_1}{s - p^*} \right\} + \mathcal{L}^{-1} \left\{ \dots \right\} = R_1 \cdot e^{p^* \cdot t}$

↓
since p^+ is a low-frequency pole $\Leftrightarrow p^+$ is "small" $\Rightarrow$
 $\Rightarrow e^{p^+ \cdot t}$ is a "slow" exponential mode

$\Downarrow$
this term in the output response can be responsible of
what is called the "Tail effect"!



⇓
by inserting a (stage) of network we give a significant "tail effect"
on the output response which could destroy the performance of the system
in terms of $\hat{S}$, t_r , t_s , $\alpha\%$

⇓
Typically the "tail effect" has relevant impact on the
settling time t_s , $\alpha\%$

